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Associations between whole blood donation and cancer incidence: a systematic review

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Abstract

Background Whole blood (WB) donation has played a critical role in the healthcare system worldwide. However, its long-term safety for donors, particularly regarding the risk of cancer, remains controversial.

Objectives We systematically reviewed the literature to examine whether WB donation is associated with altered cancer risk among donors.

Methods We searched PubMed, Scopus, Web of Science, and Google Scholar from September 1990 to December 7, 2024, without time restrictions. Only English-language publications were included. Of 4078 records screened, eleven studies met the eligibility criteria. Study quality was assessed using the Newcastle–Ottawa Scale (NOS), and findings were summarized in Tables and Figures.

Results Most studies found no significant association between WB donation and overall cancer risk once adjusted for the healthy donor effect (HDE). Reported associations varied across cancer types but were inconsistent and often limited by residual confounding.

Conclusion Current evidence does not support an increased risk of cancer among WB donors. Further well-designed studies with robust methods to address HDE are warranted to confirm these findings.

Keywords Blood donation, Whole blood, Cancer, Neoplasm, Healthy donor effect

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Introduction

Whole blood (WB) donation is critically important for the health of recipients. However, its effects on donor health are still a matter of debated. Therefore, there is a growing need for research on the safety of blood donation to reassure the donor population about their health and avoid blood supply shortages [1–3]. WB donation (and phlebotomy in general) can lead to immediate complications such as local bleeding, short-term anxiety, and syncope [4]. Apart from these effects, the short- and long-term health impacts of WB donation remain largely unexplored to date. Some studies have suggested that WB donation could lead to temporary changes in the immune system, such as increased antioxidant capacity [5] and

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decreased levels and activity of natural killer cells, which might promote the development of certain types of cancer [6–8]. The potential increase in bone marrow activity and cell proliferation following phlebotomy might itself be a risk factor for cancer [9, 10]. Conversely, it is also shown in studies that the level of iron in the human body is linked to the onset of some diseases, such as various types of cancer [6, 9]; in contrast, other studies note that declining iron levels from regular donations of even one (or a few) unit(s) per year [11] may also be a risk factor for certain diseases.

Assessing cancer risk among regular donors is complicated by biases such as the healthy donor effect (HDE), which arises from the fact that regular donors are likely a healthier subset of the population, due to their self-selection for donation and their qualification based on the criteria of the agencies they donate to (Figs. 1 and 2A) [6]. This may mask a potential adverse health effect of donation [12, 13]. As a result, it is not surprising that studies have reported conflicting findings regarding the correlation between cancer risk and regular WB donation. Some studies indicate regular donors have lower cancer risk,

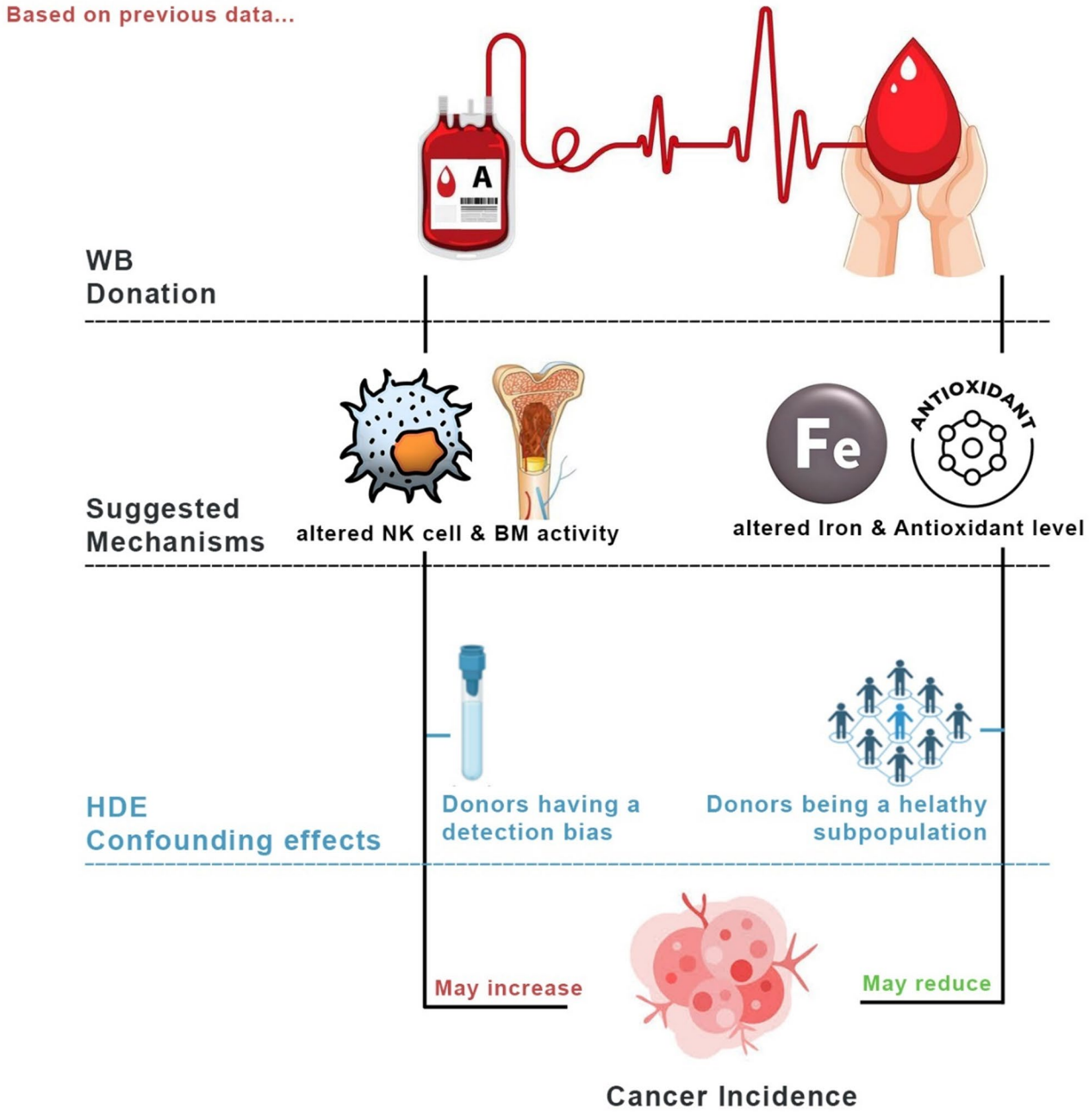


Fig. 1 Suggested mechanisms linking WB donation and cancer risk, and potential confounding by the HDE

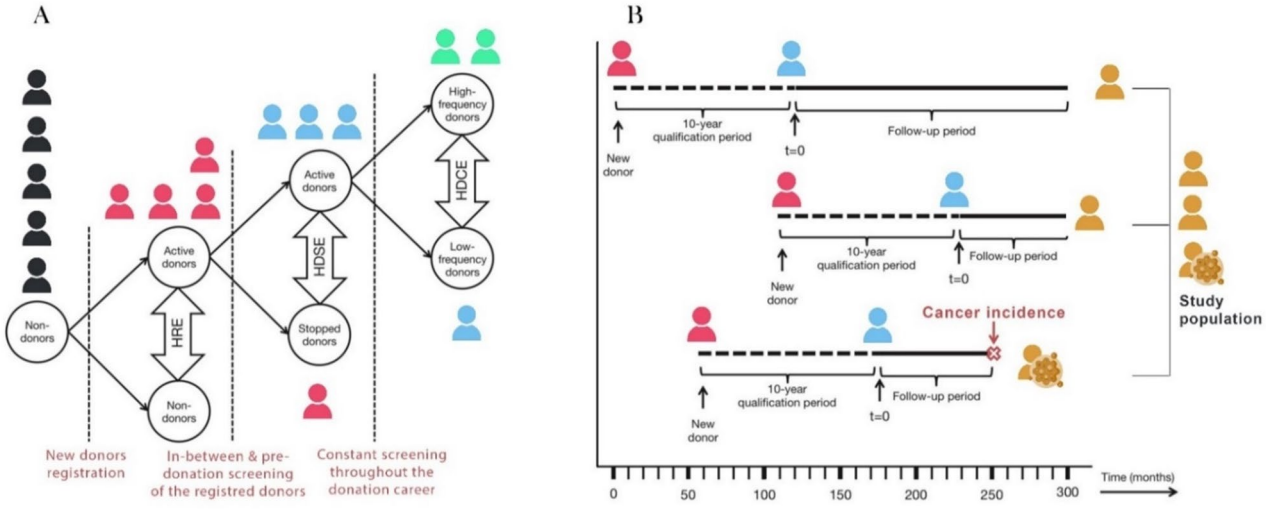


Fig. 2 Different types of HDE and the recommended "qualification period" approach to mitigate HDE. Panel **A** illustrates various levels of HDE (indicated by double arrowheads) arising from donor selection processes (dashed lines). These include healthy registration effect (HRE), healthy donor survivor effect (HDSE), and healthy donor career effect (HDCE). Panel **B** depicts the "qualification period" method, as recommended by Peffer et al. and implemented by Rahman et al., designed to reduce HDE bias. In this approach, donors undergo a 10-year qualification period to be classified as active donors before being included in the main study's follow-up phase

possibly due to their healthier lifestyles [4, 9] perhaps attributable to the HDE, while others suggest higher cancer incidence linked to increased health awareness and screening (and therefore detection) in this group [14, 15]. This confusion may cause concerns among donors and could make them skeptical about donating and hence contribute to blood supply shortages. Therefore, it's essential to investigate the health impacts of WB donation to address these concerns and prevent donors from withdrawing their donation career.

This paper provides a systematic review on cancer risk among donors by classifying methods used in past studies on the HDE and highlighting their strengths and weaknesses; synthesizing cancer-specific findings with a novel visual summary that reveals patterns not easily seen in text; and identifying methodological gaps, such as inconsistent adjustment strategies, heterogeneous donor populations, and limited qualification-period designs. These insights clarify why current evidence on cancer risk among blood donors is inconsistent and suggest directions for future research. Also, to our knowledge, no systematic review has previously addressed the correlation between WB donation and cancer incidence, prompting our study to review both older and recent literature on this association.

Materials and methods

We conducted a systematic review without meta-analysis to evaluate the risk of cancer among the blood donor population based on the current literature. The "Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)" guideline was used as the reporting protocol [16].

Literature search

We conducted a systematic search in PubMed, Scopus, Web of Science and Google Scholar. We searched for papers published from September 1990 to December 7, 2024, and focused on peer-reviewed studies in English. For PubMed, Scopus, and Web of Science, we used detailed Boolean search strings (see Appendix 1 for the full strategies). for Google Scholar we utilized keyword combinations (e.g., "Blood donation" AND "Cancer") and screened the first 200 results sorted by relevance. We excluded grey literature, trial registries, and studies that were not peer-reviewed to prioritize high quality papers (though this arises possibility of occurrence of a publication bias). All retrieved records were imported into EndNote 20.4.1 for deduplication and were manually verified for accuracy.

Eligibility criteria and data extraction

We employed the PICOS framework for study selection: (1) Population: Human participants who had donated blood. (2) Exposure: WB donation. (3) Comparator: Non-donors from the general population or lower-frequency donors. (4) Outcome: Incidence of malignant or benign neoplasms. (5) Study design: Cohort or case–control studies published in peer-reviewed journals (Table 1). We excluded case reports, reviews, conference abstracts, letters, studies lacking original data, and articles not addressing the correlation of blood donation and cancer incidence. Plasma

Table 1 Basic characteristics of the included studies

Characteristic	Number of studies
Study design	
Nested case-control in the cohort	4
Cohort	7
Country	
Sweden	5
Denmark	3
United states	3
China	1
Poland	1
Australia	1
Type of cancer	
Non-Hodgkin lymphoma	9
Polycythemia vera	9
Other hematological malignancies	8
Colorectal	8
Gastrointestinal	7
Breast	5
Other	5
Total	11*

*The sum of the numbers for characteristics does not necessarily equal 11, because some studies reported data from more than one country or on more than one type of cancer

or platelet donation studies were not included due to different physiological effects, which warrant separate systematic reviews. Data from eligible studies—including study design, population characteristics, cancer type, adjustments for the HDE and other confounders, study location, and main statistical findings—were extracted into a standardized Microsoft Excel sheet (Tables 2, 3, 4 and 5).

We did not conduct a meta-analysis due to significant heterogeneity among the studies. For instance, some did not account for HDE, or some applied various study designs to adjust for it. The cancer classification and indices also differed (e.g., standardized incidence ratios [SIR], observed-to-expected ratios [O/E], relative risks [RR], and odds ratios [OR]), along with variations in study design, participant characteristics, outcome definitions, and follow-up length.

To quantitatively assess heterogeneity among studies, we calculated the I^2 statistic for each cancer category represented in the forest plots. High I^2 values indicate substantial heterogeneity, making a pooled analysis inappropriate. Therefore, we performed a systematic narrative synthesis organized by primary cancer categories (e.g., hematological, breast, digestive) instead. We also created forest plots for visual representation of effect sizes and confidence intervals without pooled estimates.

Quality assessment

Quality assessment was conducted using the Newcastle-Ottawa Scale (NOS) [17] by two independent reviewers (AD and YM), with any disputes resolved through discussion or consultation with a third reviewer (MAMJ). All studies scored ≥ 7 stars, reflecting high methodological quality. A sensitivity analysis confirmed this robustness. A summary of NOS scores for all studies is presented in Supplementary Table 1, highlighting scores for each domain.

Study selection

The database search yielded 4078 records, leading to the exclusion of 2337 articles after primary screening and leaving 1741 records. After title and abstract screening, the full texts of 14 articles were then assessed for eligibility, and eleven studies ultimately met all inclusion criteria for the systematic review (Fig. 3).

Several studies have proposed mechanisms supporting a positive association between WB donation and cancer risk. However, the assessment of WB donation's impact on cancer risk may be confounded by the HDE, since donors generally constitute a healthier subset of the population due to self-selection and the stringent eligibility criteria applied by donation centers.

Results

Study characteristics

In total, eleven studies met the inclusion requirements [4, 6, 9, 15, 18–21], comprising nested cohort and case-control designs from various countries, including Sweden, Denmark, the U.S., China, Poland, and Australia. The studies examined risks across multiple cancer sites; findings varied significantly. Five studies showed a decreased cancer risk among blood donors, four found no association, and two reported an increased risk (Table 1). These patterns are illustrated in Fig. 4, in contrast detailed information on the included studies, including design, population, cancer sites, and quality assessment, is provided in Tables 2, 3, 4 and 5.

Overall findings

Overall, most studies ($n = 7$) showed little or no significant relationship between WB donation and cancer incidence, while five studies reported varying cancer risks—four suggested a decrease, and one indicated an increase (Tables 2, 3, 4 and 5).

Quantitative assessment using the I^2 statistic showed high heterogeneity across studies for all major cancer categories: Digestive/GI cancers (I^2 = 82%), Breast cancer (I^2 = 78%), Hematologic/lymphatic cancers (I^2 = 85%). These high I^2 values support the observed variability in study findings and reinforce our decision not to calculate pooled effect estimates.

Table 2 Studies on WB donation and risk of head and neck, digestive system, and respiratory system cancers

Type(s) of cancer	Reference	Design and population	Adjustment for HDE & detection bias	Location and years	Findings (95%-99% CI)	QA (NOS)
Head and neck	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	O/E: 0.64 (0.47–0.86)	8/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	Lip SIR: 0.71 (0.55–0.89)Pharynx SIR: 0.78 (0.66–0.91)Salivary glands SIR: 1.05 (0.81–1.35)Tongue SIR: 0.78 (0.62–0.98) Mouth SIR: 0.66 (0.54–0.81)Eye SIR: 0.96 (0.74–1.21)	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	Oral cavity and pharynx SIR in males: 1.24 (1.10–1.38), and females: 1.01 (0.60–1.60)	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	Benign neoplasm in major salivary gland RR in male: 0.86 (0.74–1.16)Benign neoplasm of mouth and pharynx RR in male: 0.92 (0.83–1.09)	9/9
Digestive system	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	Esophagus/stomach O/E: 0.54 (0.37–0.76)colorectal O/E: 0.78 (0.64–0.94)Liver O/E: 0.60 (0.39–0.88)pancreas O/E: 0.64 (0.43–0.91) (0.81 (0.54–1.47) with 10 years latency)	8/9
	Lasek et al., 1994 [19]	Cohort, 3126 donors	No	Poland, 1969–1984	Esophagus O/E: 0.9stomach O/E: 0.5colon O/E: 0.4rectum O/E: 1.1Liver O/E: 4.4pancreas O/E: 2.1	7/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	Esophagus SIR: 0.75 (0.64–0.87)stomach SIR: 0.79 (0.71–0.86)small intestine SIR: 0.80 (0.63–0.99)colon SIR: 0.94 (0.89-1.00)rectum SIR: 0.95 (0.88–1.01)Liver SIR: 0.66 (0.55–0.78)gallbladder SIR: 0.66(0.55–0.79)Liver, not specified as primary SIR: 0.58 (0.44–0.76)pancreas SIR: 0.86 (0.78–0.94) peritoneum and unspecified SIR: 0.77 (0.45–1.23)	9/9
	Edgren et al., 2008 [9]	Nested case-control study within a cohort, 118,006 donors in case-control	By restricting the analysis to donors and making dose-response comparisons among them	Sweden, 1982–2004	Cancers of the lung, liver, esophagus, stomach, and colon, with more than 5.4 g loss of iron and exposure window of 3–12 years before diagnosis OR: 0.77 (0.59–0.99)more than 2.7 g and exposure window of 3–7 years before diagnosis OR: 0.75 (0.64–0.88)no significant effect for other amounts of iron loss and other exposure windows	9/9
	Zhang et al., 2012 [21]	Cohort, 257 donors and 34,864 non-donors	By investigating dose-response effects and utilizing the Health Professionals Follow-up Study (HPFS)	U.S, 1992–2008	Colorectal, colon, rectal, proximal colon, distal colon RR in male: No significant association found	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	SIR in males: 1.24 (1.10–1.38), and females: 1.67 (1.45–1.90)	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	esophagus RR: male: 0.55 (0.41–0.72), female: 0.73 (0.62–0.88)stomach RR in male: 0.92 (0.85–1.09), and female: 1.14 (0.91–1.34)colorectum RR in male: 0.96, and female: 1.22Liver RR in males: 0.42 (0.28–0.67), and females: 0.57 (0.42–0.79)pancreas RR in male: 0.87 (0.75–1.04)neoplasms of uncertain behavior of digestive organs RR:0.89 (0.75–1.27)benign neoplasms of ill-defined parts of the digestive system RR: 0.85 (0.64–1.02)	9/9

Table 2 (continued)

Type(s) of cancer	Reference	Design and population	Adjustment for HDE & detection bias	Location and years	Findings (95%-99% CI)	QA (NOS)
	Rahman et al., 2024 [6]	Cohort, 2867 regular high-frequency donors and 3888 low-frequency donors	by using the qualification period method, restricting the population to low and high-frequency donors and investigating dose-response effects, separating the exposure period and follow-up period and applying several statistical adjustments in the analyses	Australia, 2006–2020	gastrointestinal/colorectal, 5-year risk difference RR: 0.2%	9/9
Respiratory system	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	Lung O/E: 0.61 (0.50–0.74)	8/9
	Lasek et al., 1994 [19]	Cohort, 3126 donors	No	Poland, 1969–1984	Larynx O/E: 0.6lung O/E: 1.1	7/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	Nasal cavities and sinuses SIR: 0.84 (0.59–1.17)larynx SIR: 0.68 (0.57–0.80)lung primary, trachea SIR: 0.77 (0.73–0.81) pleura SIR: 1.06 (0.84–1.31)lung, not specified as primary SIR: 0.85 (0.56–1.25)mediastinum SIR: 1.68 (0.95–2.72)	9/9
	Edgren et al., 2008 [9]	Nested case–control study within a cohort, 118,006 donors in case-control	By restricting the analysis to donors and making dose-response comparisons among them	Sweden, 1982–2004	Cancers of the lung, liver, esophagus, stomach, and colon, with more than 5.4 g loss of iron and exposure window of 3–12 years before diagnosis OR: 0.77 (0.59–0.99)more than 2.7 g and exposure window of 3–7 years before diagnosis OR: 0.75 (0.64–0.88)no significant effect for other amounts of iron loss and other exposure windows	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	SIR in male: 0.93(0.79–1.08), and female: 1.48(1.25–1.74)	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	Lung RR in male: 0.74(0.59–0.88), and female: 0.95(0.66–1.16)	9/9

Abbreviation: *QA* Quality Assessment

Site-specific patterns

Below, we summarize the main patterns reported for major cancer groups; complete study-level data and effect estimates are available in Tables 2, 3, 4 and 5.

Head and neck cancers

Four cohort studies reported on head and neck sites [4, 15, 18, 20]. Two studies [18, 20] reported decreased risks (but did not adjust for HDE, one [15] reported no significant association, and one [4] noted increased incidence in males (the authors of the latter study discussed the possibility of detection bias) (Table 2 for details).

Digestive/gastrointestinal cancers

Eight studies (seven cohorts and one case–control) evaluated gastrointestinal sites [4, 6, 9, 15, 18–21]. Three studies [6, 9, 21] that explicitly controlled for donor selection bias found little or no association (Fig. 5). The nested case–control study [9] that assessed iron loss (dose–response) reported a lower combined risk for several iron-related cancers in men with high iron loss. Some studies reported decreased risks [4, 18, 20], while one smaller cohort reported an increased risk for liver cancer [19], and one reported an increased risk in both genders [15] (Table 2; Fig. 1).

Respiratory cancers

Six studies examined respiratory-site cancers (five cohort studies and one case-control study) [4, 9, 15, 18–20]. A dose–response analysis reported reduced odds for respiratory and digestive cancers in high iron-loss donors [9], while others had varying results or detection bias concerns [18] (Table 2).

Table 3 Studies on WB donation and risk of breast, reproductive system, urinary system, and skin cancers

Type(s) of cancer	Reference	Design and population	Adjustment for HDE & detection bias	Location and years	Findings(95–99%CI)	QA (NOS)
Breast	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	O/E: 1.03 (0.85–1.22) (1.23 (1.06–1.91) with 10 years latency)	8/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	SIR: 1.08 (1.04–1.12)	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	SIR in females: 1.79 (1.66–1.92)	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	malignant RR in females: 0.91 (0.81–1.03)benign breast tumor RR in females: 0.93 (0.84–1.06)	9/9
	Rahman et al., 2024 [6]	Cohort, 2867 regular high-frequency donors and 3,888 low-frequency donors	by using the qualification period method, restricting the population to low- and high-frequency donors and investigating dose-response effects, separating the exposure period and follow-up period, and applying several statistical adjustments in the analyses	Australia, 2006–2020	5-year RR: -0.2%	9/9
Reproductive system	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	Female gonads/genitalia O/E: 0.83 (0.66–1.06)male gonads/genitalia O/E: 0.61 (0.36–0.98) (0.78(0.24–2.24) with 10 years latency)prostate: 0.96 (0.81–1.14)	8/9
	Lasek et al., 1994 [19]	Cohort, 3126 donors	No	Poland, 1969–1984	Testis O/E: 2prostate O/E: 2.3	7/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	cervix uteri SIR: 0.77 (0.69–0.85)corpus uteri SIR: 0.83 (0.75–0.92)uterus, other parts, and unspecified SIR: 0.76 (0.55–1.03)ovary, fallopian tube, broad ligament SIR: 0.93 (0.85–1.02)other and unspecified female genital organs SIR: 0.79 (0.57–1.05)prostate SIR: 1.21 (1.16–1.26)testis SIR: 1.20 (1.09–1.32)other and unspecified male genital organs SIR: 0.86	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	Male genital SIR: 2.48(2.32 2.63)female genital SIR: 1.50(1.32 1.70)	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	prostate RR: 0.44 (0.30–0.66)ovary RR: 0.97 (0.75–1.26), uterus RR: 0.88 (0.75–1.03), cervix RR: 1.03 (0.90–1.10)benign ovarian tumor RR: 0.94 (0.74–1.16)carcinoma in situ of cervix RR: 0.96 (0.85–1.09)benign tumors in the female reproductive system RR: 0.92 (0.85–1.09)uterine leiomyoma RR: 0.85 (0.71–1.12)other benign tumors of the uterus RR: 0.97 (0.93–1.01)	9/9
Urinary system	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	Urinary organs O/E: 0.78 (0.64–0.94)	8/9
	Lasek et al., 1994 [19]	Cohort, 3126 donors	No	Poland, 1969–1984	bladder O/E: 0.6, kidney O/E: 0.4	7/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	Kidney SIR: 0.84 (0.77–0.91)bladder including papilloma SIR: 0.98 (0.92–1.04)	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	urinary SIR in male: 1.15 (0.97 1.37), and female: 1.16 (0.84 1.55)	9/9

Table 3 (continued)

Type(s) of cancer	Reference	Design and population	Adjustment for HDE & detection bias	Location and years	Findings(95–99%CI)	QA (NOS)
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	bladder RR in male: 1.03 (0.85–1.25)	9/9
Skin	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	Malignant melanoma of skin O/E: 1 (0.78–1.29)	8/9
	Lasek et al., 1994 [19]	Cohort, 3126 donors	No	Poland, 1969–1984	Melanoma (malignant) O/E: 0.8skin: 1.8	7/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	melanoma of skin SIR: 1.17 (1.11–1.24)other skin cancers SIR: 1.02 (0.97–1.07)	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	SIR in male: 2.19(1.89–2.53), and female: 1.77(1.48 2.11)	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	Benign lipoid tumor RR in males: 0.97 (0.92–1.05), and females: 0.88 (0.72–1.24)other benign neoplasms of skin RR in male: 0.79 (0.62–0.94)	9/9

- **Hematologic Lymphatic Cancers**: Several studies did not find significant associations,

Breast cancer

Five cohort studies assessed breast cancer [4, 6, 15, 18, 20]. Mixed findings were observed across five cohort studies, with one reporting no change in risk after mitigating confounding factors (Table 3; Fig. 6).

Reproductive, urinary, and other site-specific cancers

Results for reproductive and urinary tract cancers were inconsistent [4, 15, 18–20], with most studies showing null findings for skin [4, 15, 18–20], brain/nervous system [4, 15, 18–20], and other cancers [4, 9, 15, 18–20] (Tables 3 and 5).

Hematologic/lymphatic cancers

Several studies with well-controlled designs reported no significant associations [6, 7, 9, 14, 15, 22]. While older or less controlled studies showed mixed results [4, 18–20]. (Table 4; Fig. 7). Detailed data is available in the supplementary Tables 1 and 2.

Discussion

The safety of WB donation for donors warrants careful investigation, particularly given the emerging evidence suggesting a potential impact on donor health, despite the life-saving role of blood donation for recipients [23]. Considering the inadequacy of volunteer numbers and the common problem of WB storage shortage, research that reinforces the safety of blood donation could also help support donor recruitment and retention [24]. Previously, some studies have reported increased risks for certain cancers among blood donors compared to the general population (see Tables 2, 3, 4 and 5). To our knowledge, this review — the first of its kind — conducted a systematic search across four major databases to identify peer-reviewed studies assessing cancer risk in relation to WB donation. A key methodological concern in the current literature is the insufficient adjustment for the HDE, a form of selection bias whereby blood donors represent a healthier subpopulation due to regular health screening. Most studies that reported significant associations, whether positive or negative, did not adequately control for HDE. Some authors attempted to mitigate this bias by comparing high-frequency or long-career donors with their lower-frequency counterparts. Others limited inclusion to donors with ≥ 5 years of donation history (the exposure window/qualification period method), or comparing survival among cancer-diagnosed donors with matched non-donor controls. None of these studies reported a robust association between blood donation and cancer risk that was free from possible detection bias or residual confounding by HDE.

Study designs and methodologies varied substantially across studies, which likely contributed to the inconsistent findings. A significant source of this heterogeneity was the different approaches used to mitigate the HDE. Of the eleven studies included, four did not apply any explicit adjustment for HDE, three of which reported statistically significant associations in either direction. In contrast, most studies that attempted to address HDE reported no clear association between WB donation and cancer risk. Differences in confounder adjustment and cancer outcome classification further limited comparability across studies. In addition, donor data were derived from different blood collection systems with distinct eligibility criteria, which may have contributed to residual confounding.

Table 4 Studies on WB donation and risk of Brain and nervous system, Thyroid, Hematological/ lymphatic, and Endocrine gland cancers. (Abbreviations: HR, hazard ratio)

Type(s) of cancer	Reference	Design and population	Adjustment for HDE & detection bias	Location and years	Findings(95–99%CI)	QA (NOS)
Brain and nervous system	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	Nervous system O/E: 0.73(0.53–0.97) (0.73(0.28–1.07) with 10 years latency)	8/9
	Lasek et al., 1994 [19]	Cohort, 3126 donors	No	Poland, 1969–1984	brain O/E: 1.3	7/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	SIR:0.94 (0.89-1.00)	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	SIR in male: 1.55(1.11 2.10), and female: 1.59(1.05 2.31)	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	brain RR in male: 1.19 (1.06–1.29) meningioma RR in female: 1.03 (0.92–1.07)	9/9
Thyroid	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	O/E: 1.27 (0.74–2.03)	8/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	SIR: 1.05 (0.92–1.19)	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	Thyroid malignant RR in females: 0.85(0.72–1.01)benign thyroid tumor RR in male: 0.95 (0.89–1.05)	9/9
Hematological/ lymphatic	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	Hematological O/E: 0.89 (0.73–1.08)	8/9
	Lasek et al., 1994 [19]	Cohort, 3126 donors	No	Poland, 1969–1984	Lymphatic and hemopoietic tissue O/E: 0.9	7/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	NHL SIR: 0.96 (0.89–1.03)HL SIR: 0.98 (0.84–1.13)MM SIR: 0.91 (0.80–1.03) Leukemia SIR: 0.97 (0.89–1.06)	9/9
	Edgren et al., 2008 [9]	Nested case–control study within a cohort, 118,006 donors in case-control	By restricting the analysis to donors and making dose-response comparisons among them	Sweden, 1982–2004	NHL and other hematologic/ lymphatic malignancies OR: no significant effect for WB donation	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	lymphoma SIR in male: 1.42(1.18–1.69), and female: 1.72(1.39–2.12) myeloma SIR in male: 2.11(1.45–2.96), and female: 1.79(1.02–2.90) leukemia SIR in male: 2.14(1.70–2.66), and female: 1.33(0.88 1.92)	9/9
	Ishii et al., 2016 [7]	Cohort, 36,576 donors	By implementing HPFS, which is comprised entirely of U.S. health professionals, investigating the dose-response effect, and having access to detailed, high-quality information on lifestyle	U.S, 1992–2010	All NHL, DLBCL, FL, CLL/LL HR: no significant association found.	9/9
	Edgren et al., 2016 [22]	nested case-control, 2637 donors	By investigating a dose-response effect and applying several statistical adjustments	Sweden, 1980–2012	No significant effect was found.	9/9
	Zhao et al., 2020 [14]	nested case-control, 1,021,433 donors	By investigating a dose-response effect and including controls with at least one donation	Sweden, 1968–2012	ALL, AML, CLL, CML, HL, Myeloma, and other NHL: no significant effect was found	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	lymphodia RR in male: 0.75(0.62–0.85)hemangioma and lymphangioma RR in male: 0.75 (0.51–0.89), and female: 0.65(0.44–0.83)	9/9

Table 4 (continued)

Type(s) of cancer	Reference	Design and population	Adjustment for HDE & detection bias	Location and years	Findings(95–99%CI)	QA (NOS)
	Rahman et al., 2024 [6]	Cohort, 2867 regular high-frequency donors and 3,888 low-frequency donors	by using the qualification period method, restricting the population to low- and high-frequency donors and investigating dose-response effects, separating the exposure period and follow-up period, and applying several statistical adjustments in the analysis	Australia, 2006–2020	5-year RR: 0%	9/9
Endocrine glands	Edgren et al., 2007 [20]	Nested case–control study within a cohort, 118,006 donors in case-control	No	Sweden, 1967–2002	SIR: 0.92 (0.83–1.03)	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	SIR in male: 0.78(0.47–1.21), and female: 0.91(0.70–1.16)	9/9

Table 5 Studies on WB donation and of miscellaneous and overall cancer risk

Type(s) of cancer	Reference	Design and population	Adjustment for HDE & detection bias	Location and years	Findings(95–99%CI)	QA (NOS)
Other cancer	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	Benign neoplasm of other and unspecific endocrine glands RR in female: 1.10 (0.86–1.27)	9/9
	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	Miscellaneous O/E: 0.83 (0.67–1.01)	8/9
	Lasek et al., 1994 [19]	Cohort, 3126 donors	No	Poland, 1969–1984	other sites O/E: 0.6	7/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	Bone SIR: 1.02 (0.77–1.32)connective tissue SIR: 1.02 (0.88–1.19)metastases SIR: 0.72 (0.54–0.94) other and unspecified sites SIR: 0.77 (0.69–0.84)	9/9
	Edgren et al., 2008 [9]	Nested case–control study within a cohort, 118,006 donors in case-control	By restricting the analysis to donors and making dose-response comparisons among them	Sweden, 1982–2004	Solid cancer OR: no significant association found	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	Other SIR in male: 1.45(1.16–1.78), and female: 1.68(1.23–2.24)	9/9
	Su et al., 2022 [4]	Cohort, 1,625,599 donors and 1,625,599 non-donors	No	China, 2012–2018	Benign tumors in the anus and anal canal RR in male: 0.87 (0.78–1.28), benign neoplasm of bone and articular cartilage RR in male: 1.05 (0.89–1.15)	9/9
	Merk et al., 1990 [18]	Cohort, 36,795 donors	No	Sweden, 1970–1986	O/E: 0.89 (0.73–1.08) (0.98 (0.76–1.23) with 10 years latency)	8/9
All sites	Lasek et al., 1994 [19]	Cohort, 3126 donors	No	Poland, 1969–1984	O/E: 1.0 (57 observed / 54.4 expected)	7/9
	Edgren et al., 2007 [20]	Cohort, 1,110,329 donors	No	Sweden, 1967–2002	SIR: 0.96 (0.89–1.03)	9/9
	Vahidnia et al., 2013 [15]	Cohort, 55,158 donors	By comparing survival in donors to a matched sample of non-donors	U.S, 1982–2002	SIR in males: 1.68 (1.61–1.75), and females: 1.60 (1.53–1.68) (however, the mortality of donor cancer patients was significantly less; therefore, the authors addressed the possible effect of detection bias)	9/9

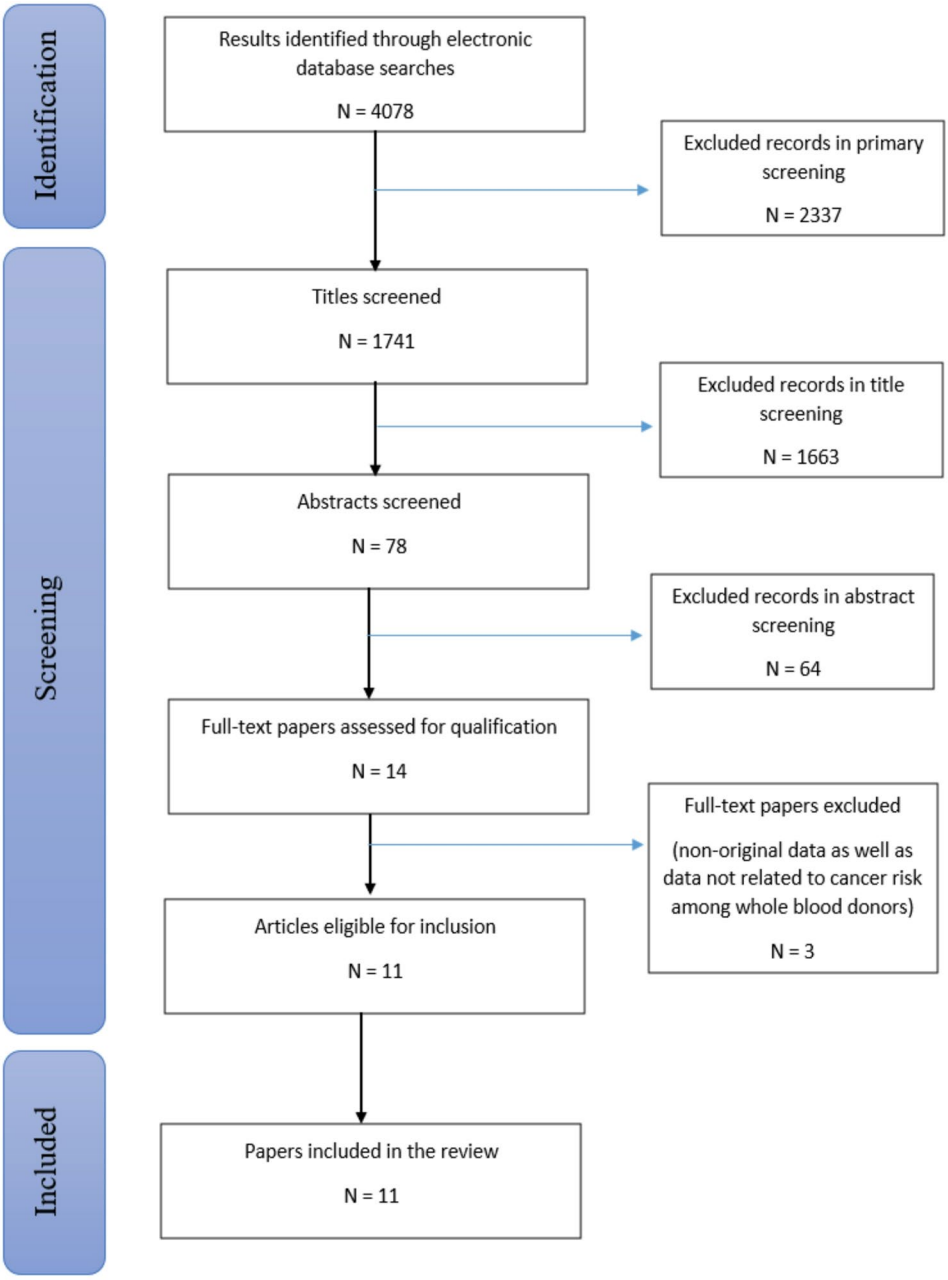


Fig. 3 Schematic diagram of the search strategy, screening criteria, eligibility criteria, and inclusion criteria (PRISMA guideline)

Beyond methodological limitations, it is essential to consider biological plausibility. Repeated WB donation induces physiological changes that could exert both protective and potentially deleterious effects on cancer risk. On the protective side, regular phlebotomy reduces iron stores, thereby decreasing iron-driven oxidative stress and free radical–mediated DNA damage [25]. Blood donation has also been associated with reduced levels of acute-phase reactants, leading to improved antioxidant capacity and potentially lowering systemic inflammation [26]. Conversely, repeated WB donation might suppress certain aspects of immune surveillance: for instance, decreased natural killer (NK) cell activity could impair the immune system's ability to detect and eliminate malignant cells [8], while increased bone marrow activity required to compensate for repeated cell loss could theoretically raise the risk of mutations during accelerated hematopoiesis [9]. Although these mechanisms remain largely theoretical in the donor context, they provide a helpful framework for interpreting conflicting epidemiological findings.

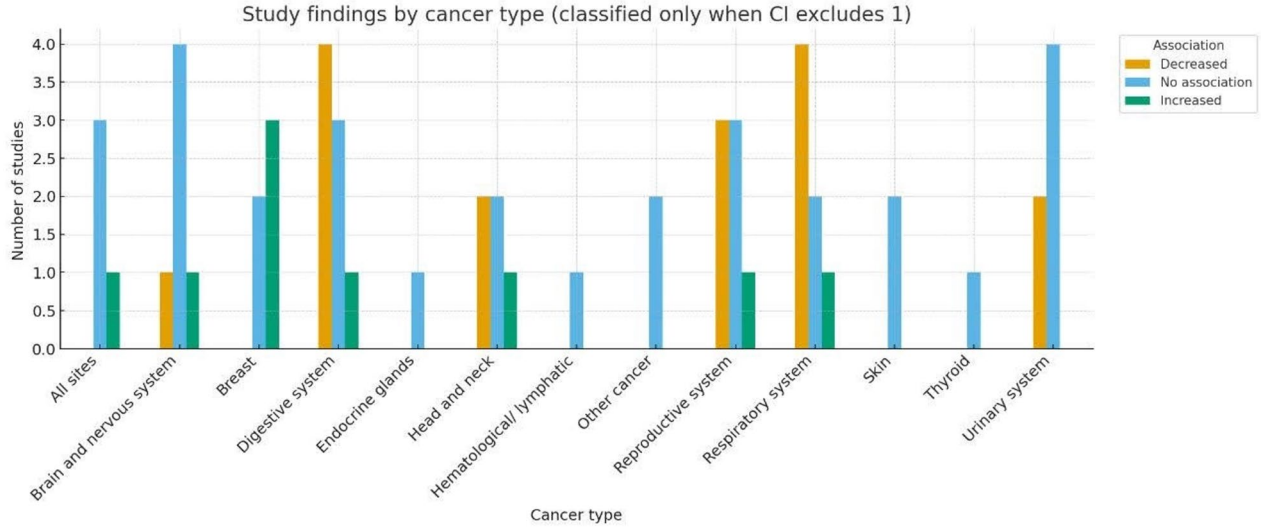


Fig. 4 Distribution of study findings by cancer type regarding WB donation and cancer risk. Associations were classified as "decreased" or "increased" only when effect estimates had 95% confidence intervals excluding 1. All other findings (including those without reported 95% CIs or with 95% CIs crossing 1) were categorized as "no association."

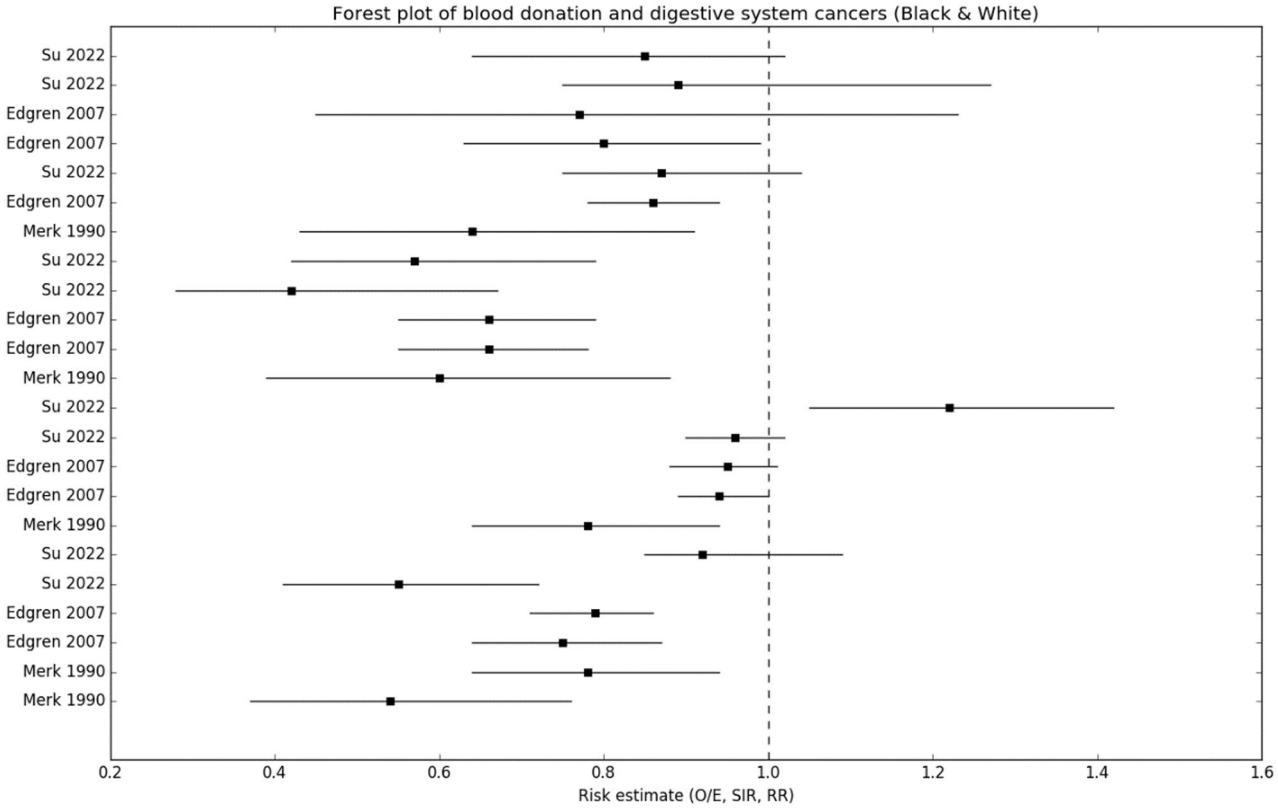


Fig. 5 Forest plot of studies on WB donation and the risk of digestive/gastrointestinal cancers (O/E, SIR, RR, HR, OR with 95% CI)

It is also notable that the current evidence base focuses predominantly on WB donation. In contrast, plasma and platelet apheresis involve different physiological stresses. For example, plasma donation is associated with repeated exposure to citrate, which can influence calcium homeostasis and metabolic pathways [27], whereas platelet donation may have distinct immunomodulatory effects. A more precise comparison of WB donors with apheresis donors could help clarify whether observed risk

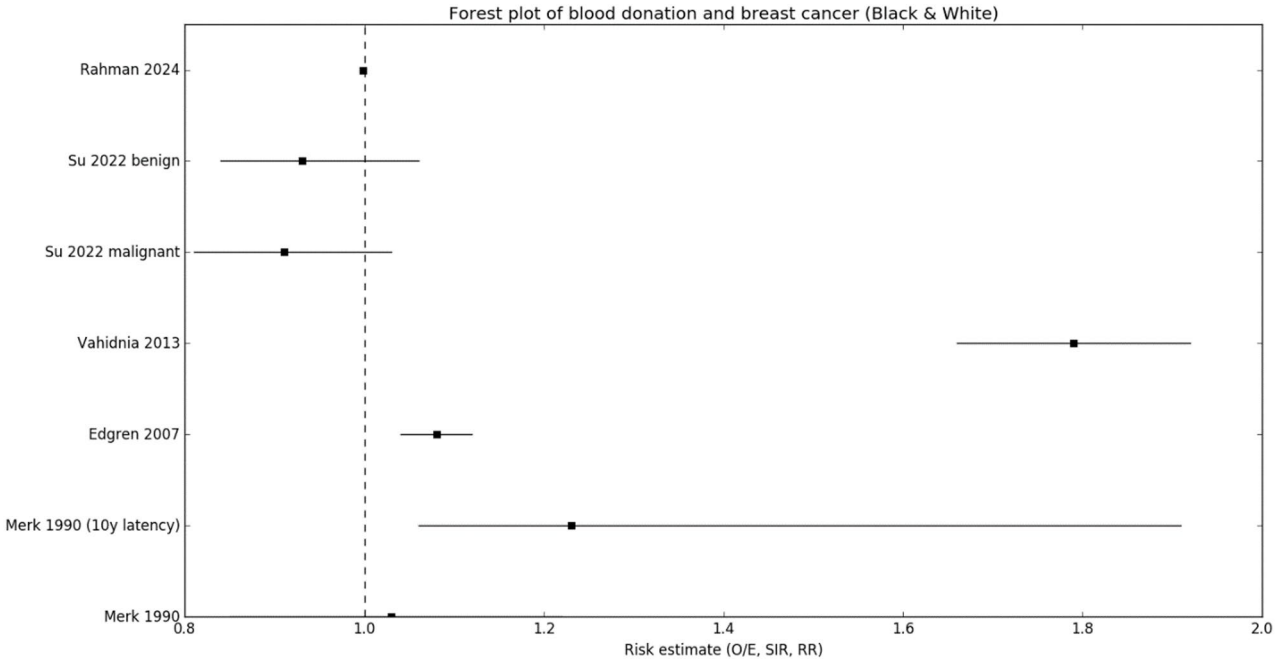


Fig. 6 Forest plot of studies on WB donation and the risk of breast cancer (O/E, SIR, RR, HR, OR with 95% CI)

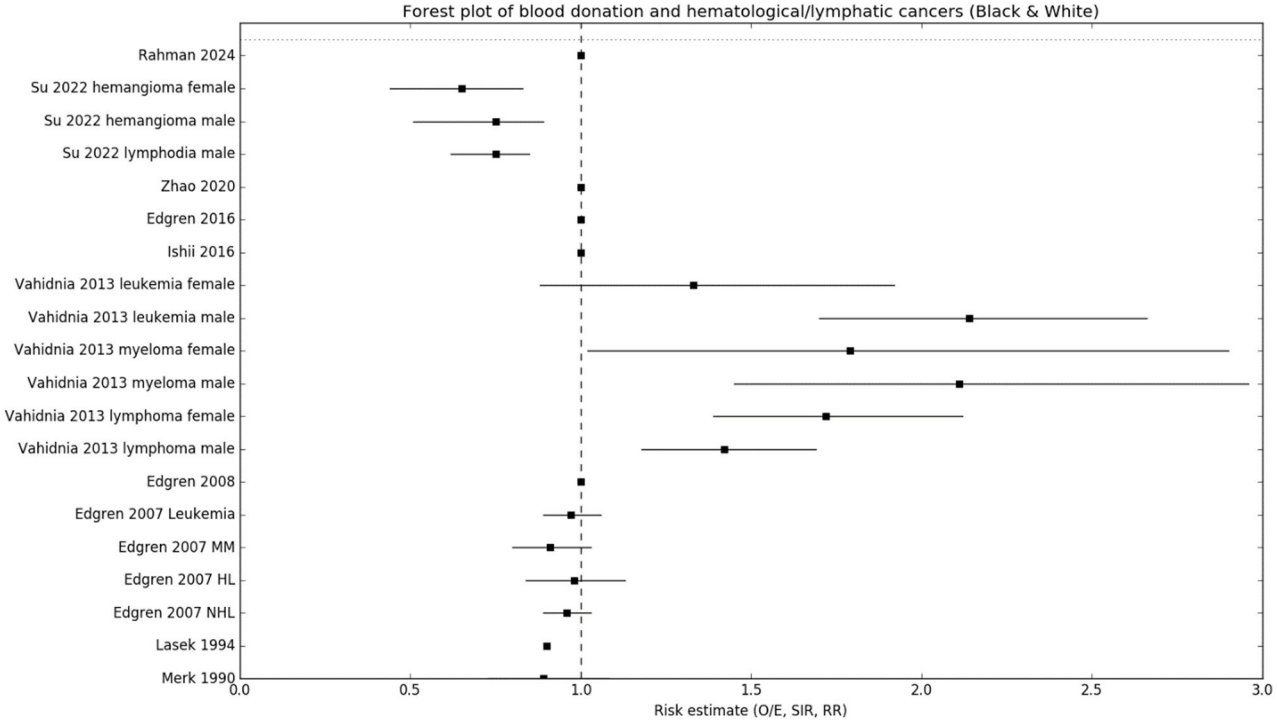


Fig. 7 Forest plot of studies on WB donation and the risk of hematologic/lymphatic cancer (O/E, SIR, RR, HR, OR with 95% CI)

patterns are unique to WB donation or reflect broader consequences of repeated blood component removal.

Taken together, the current evidence does not suggest a consistent increase in cancer risk among WB donors. However, substantial heterogeneity across studies and persistent residual confounding, particularly related to the HDE, limit causal interpretation. Future research should prioritize robust study designs that more effectively mitigate selection bias and evaluate donor populations across diverse geographical settings to clarify the long-term relationship between WB donation and cancer risk.

Limitations

The findings of this systematic review should be interpreted in light of several significant limitations. First, the number of available studies examining the association between WB donation and cancer risk remains limited, with only eleven studies meeting the inclusion criteria. A substantial proportion of these studies were published before 2010, which may limit their applicability to contemporary donor populations and donation practices.

Second, considerable heterogeneity was observed across studies in terms of design, exposure definitions, outcome classification, and analytical approaches. In particular, methods used to address the HDE varied widely, and several studies did not apply any explicit adjustment for this bias. As a result, residual confounding related to donor selection and health status cannot be excluded. Differences in confounder adjustment, cancer classification, and reporting metrics further limited direct comparability between studies.

Due to this heterogeneity and the methodological diversity of the included studies, a quantitative meta-analysis was not feasible. Variability in effect measures, inconsistent categorization of cancer outcomes, and differences in exposure assessment precluded meaningful data pooling. Consequently, findings were synthesized narratively.

The geographical scope of the included studies represents another limitation. Most studies were conducted in a small number of high-income countries, primarily in Northern Europe and North America. This restricts the generalizability of the findings to donor populations in other regions with different healthcare systems, eligibility criteria, and demographic characteristics.

Additionally, this review was limited to peer-reviewed observational studies published in English. While this approach ensured methodological quality and interpretability, it may have resulted in the exclusion of relevant unpublished or non-English studies, raising the possibility of publication bias (although our database searches did not yield any relevant non-English studies). Smaller studies with null findings may therefore be underrepresented.

Finally, the evidence reviewed focused exclusively on WB donation. The potential association between other donation types, such as plasma or platelet donation, and cancer risk was not addressed and remains an important area for future research.

None of the studies applied a restriction to include only donors with a donation career of at least ten years (Fig. 2B), as recommended by Peffer et al. [28], to minimize HDE. Furthermore, as some of the included studies relied on registry-based data, the possibility of misclassification or incomplete recording of donor characteristics and cancer diagnoses must also be mentioned.

Another limitation is that most included studies did not consistently report sex-specific analysis, and employed aggregate rather than individual-level data on frequency or duration of blood donation as a time-dependent exposure. They also mostly applied varying definitions of cancer outcomes and donor eligibility criteria by nation. Follow-up in a few cases may have been too brief to encompass the lengthy latency of cancer onset.

Finally, differences in registry coverage, no data on lifestyle-related confounders, exclusion of grey literature, and the vast time span of the included studies (from the early 1990s to 2024) also make comparability more difficult. The absence of comprehensive sensitivity or subgroup analyses in most of the studies further weakens the validity of their conclusions.

Conclusion

Conclusively, the available evidence does not indicate an increase in cancer risk among WB donors. However, interpretation of existing findings is limited by substantial heterogeneity across studies and persistent methodological challenges, particularly inadequate control for the HDE. Differences in study design, confounder adjustment, and donor selection further complicate cross-study comparisons.

While the overall evidence base has expanded over the past decade, the current literature remains insufficient to support a definite association between cancer risk and WB donation. Future studies employing robust methodological approaches are needed to better clarify whether cancer risk has any associations with WB donation. Such evidence is essential to ensure donor safety and maintain donation practices and consequent health outcomes.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

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Authors' contributions

Yasaman Moghaddasi, Vahid Rahmanian, and Mirza Ali Mofazzal Jahromi designed this investigation. Yasaman Moghaddasi and Abbas Dehghanian extracted data, wrote and paraphrased the paper. Abbas Dehghanian and Yasaman Moghaddasi provided the tables and figures. Finally, Armaghan Vafafar, Vahid Rahmanian, and Mirza Ali Mofazzal Jahromi critically edited this paper.

Data availability

Raw data were generated at Jahrom University of Medical Sciences during this study. Derived data supporting the findings of this study are available from the corresponding author on request.

Declarations

Ethics approval and consent to participate
This study was performed according to the guidelines of the Medical Ethics Committee of the Jahrom University of Medical Sciences (IR.JUMS.REC.1403.072). The authors declare that ChatGPT version 5.2 (OpenAI, USA) and Grammarly (Grammarly, USA) was used for paraphrasing and editing the final context based on BMC policy.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

1. Rezaei S, Amiri F, Delangizan S. Discrete choice experiment to determine preferences of blood donation in Iran. Sci Rep. 2025;15(1):14947.
2. Reiss RF. Blood donor well-being: a primary responsibility of blood collection agencies. Ann Clin Lab Sci. 2011;41(1):3–7.
3. Infanti L. Protecting the blood donor: ferritin-based intervals to improve donor health. Lancet. 2024;404(10447):2–3.
4. Su S, Ma T, Sun Y, Guo L, Su X, Wang W, et al. Association between blood donation and malignant and benign tumour risk: A population-based study of 3.4 million participants in China. J Oncol. 2022;2022:7647431.
5. Yunce M, Erdamar H, Bayram NA, Gok S. One more health benefit of blood donation: reduces acute-phase reactants, oxidants and increases antioxidant capacity. J Basic Clin Physiol Pharmacol. 2016;27(6):653–7.
6. Rahman MM, Hayen A, Olynyk JK, Cust AE, Irving DO, Karki S. Regular whole blood donation and gastrointestinal, breast, colorectal and haematological cancer risk among blood donors in Australia. Vox Sang. 2024.
7. Ishii K, Birmann BM, Zhang X, Giovannucci E, Bertrand KA. A prospective analysis of blood donation history and risk of non-Hodgkin lymphoma. Leuk Lymphoma. 2016;57(6):1423–8.
8. Marquet RL, van Hoynck MA, Busch OR, Jeekel J. Blood donation leads to a decrease in natural killer cell activity: a study in normal blood donors and cancer patients. Transfusion. 1993;33(5):368–73.
9. Edgren G, Reilly M, Hjalgrim H, Tran TN, Rostgaard K, Adami J, et al. Donation frequency, iron loss, and risk of cancer among blood donors. J Natl Cancer Inst. 2008;100(8):572–9.
10. Frangoul H, Navarro WH, Confer DL, Jagasia M. Infusion of autologous peripheral blood stem cells in an unrelated donor who developed severe aplastic anemia following stem cell donation. Bone Marrow Transplant. 2012;47(6):869.
11. Mantadakis E, Panagopoulou P, Kontekaki E, Bezirgiannidou Z, Martinis G. Iron deficiency and blood donation: Links, risks and management. J Blood Med. 2022;13:775–86.
12. van den Hurk K, Zalpuri S, Prinsze FJ, Merz EM, de Kort W. Associations of health status with subsequent blood donor behavior-An alternative perspective on the healthy donor effect from donor insight. PLoS ONE. 2017;12(10):e0186662.
13. Su S, Sun Y, Gu X, Wu W, Su X, Ma T, et al. Exploration of the healthy donor effect among 0.6 million blood donors in china: longitudinal study. JMIR Public Health Surveill. 2024;10:e48617.
14. Zhao J, Dahlén T, Brynolf A, Edgren G. Risk of hematological malignancy in blood donors: a nationwide cohort study. Transfusion. 2020;60(11):2591–6.
15. Vahidnia F, Hirschler NV, Agapova M, Chinn A, Busch MP, Custer B. Cancer incidence and mortality in a cohort of US blood donors: a 20-year study. J Cancer Epidemiol. 2013;2013(1):814842.
16. Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. Syst Reviews. 2015;4:1–9.
17. Wells GA, Shea B, O'Connell D, Peterson J, Welch V, Losos M et al. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. 2000. https://ohri.ca/en/who-we-are/core-facilities-and-platforms/ottawa-methods-centre/newcastle-ottawa-scale.
18. Merk K, Mattsson B, Mattsson A, Holm G, Gullbring B, Björkholm M. The incidence of cancer among blood donors. Int J Epidemiol. 1990;19(3):505–9.
19. Lasek W, Jakobisiak M, Stoklosa T. Decreased natural-killer-cell activity in whole-blood donors does not seem to result in increased cancer incidence. Transfusion. 1994;34(4):359–60.
20. Edgren G, Tran TN, Hjalgrim H, Rostgaard K, Shanwell A, Titlestad K, et al. Improving health profile of blood donors as a consequence of transfusion safety efforts. Transfusion. 2007;47(11):2017–24.
21. Zhang X, Ma J, Wu K, Chan AT, Fuchs CS, Giovannucci EL. Blood donation and colorectal cancer incidence and mortality in men. PLoS ONE. 2012;7(6):e39319.
22. Edgren G, Nyrén O, Hultcrantz M, Nielsen KR, Pedersen OBV, Björkholm M, et al. Blood donation and risk of polycythemia Vera. Transfusion. 2016;56(6 Pt 2):1622–7.
23. France CR, France JL, Ysidron DW, Martin CD, Duffy L, Kessler DA, et al. Blood donation motivators and barriers reported by young, first-time whole blood donors: examining the association of reported motivators and barriers with subsequent donation behavior and potential sex, race, and ethnic group differences. Transfusion. 2022;62(12):2539–54.
24. Brunson DC, Belanger GA, Sussmann H, Fine AM, Pandey S, Pham TD. Factors associated with first-time and repeat blood donation: adverse reactions and effects on donor behavior. Transfusion. 2022;62(6):1269–79.
25. Houschyar KS, Lüdtke R, Dobos GJ, Kalus U, Broecker-Preuss M, Rampp T, et al. Effects of phlebotomy-induced reduction of body iron stores on metabolic syndrome: results from a randomized clinical trial. BMC Med. 2012;10:54.
26. Lainé F, Reymann JM, Morel F, Langouët S, Perrin M, Guillygomarc'h A, et al. Effects of phlebotomy therapy on cytochrome P450 2e1 activity and oxidative stress markers in dysmetabolic iron overload syndrome: a randomized trial. Aliment Pharmacol Ther. 2006;24(8):1207–13.
27. Lokhande T, Thomas S, Kumar G, Bajpai M. The play of citrate infusion with calcium in plateletpheresis donors. Indian J Hematol Blood Transfus. 2021;37(2):295–301.
28. Peffer K. Blood donation and cardiovascular disease. Addressing the healthy donor effect. SI: sn; 2015.

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